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Abstract

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Declaration

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Chapter 1

Introduction

Intelligent agents that act in alignment with human intentions are crucial for safety and usability in real-world applications. In complex environments such as robotics, healthcare, and video games, reward functions are often underspecified or difficult to handcraft.¹ Instead, it is natural to learn these reward functions from expert demonstrations, a problem formalised as Inverse Reinforcement Learning (IRL) [9].

However, standard IRL approaches struggle under conditions of sparse or suboptimal data, environment stochasticity, and latent structure in expert behaviour. In many real-world settings, demonstrated actions are influenced by unobserved variables such as social cues, risk preferences, or task context which may act as confounders.² Without modeling these causal factors, IRL can recover misleading rewards that reflect surface-level behaviour rather than underlying human intent [4].

While imitation learning approaches such as Behavioural Cloning or Generative Adversarial Imitation Learning (GAIL) aim to learn policies that replicate expert behaviour, our work focuses instead on recovering the reward function that underlies expert decision-making. This distinction is central to our framing: whereas policy learning methods imitate *what* the expert does, reward learning seeks to understand *why* the expert acts. Reward functions offer greater interpretability, transferability, and alignment with long-term goals, enabling agents to generalise to novel states and environments. In contrast, policies often fail under distribution shift or in unseen scenarios where expert behaviour was never observed.

This dissertation explores how incorporating causal assumptions into IRL can

¹In many real-world tasks, e.g., autonomous driving or assistive robotics, reward signals such as "safety" or "comfort" are hard to express explicitly but easily demonstrated by humans.

²A confounder is a latent variable that influences both expert actions and outcomes. For instance, an expert may avoid certain regions due to unseen long-term risks, not low immediate reward.

improve the alignment, robustness, and generalisation of learned reward functions. We build on recent work in causal imitation learning [2, 12] and causal reinforcement learning [1] to formulate and evaluate a *Causal Deep Inverse Reinforcement Learning* (Causal-DIRL) framework. Our hypothesis is that explicitly modelling structural dependencies behind expert demonstrations enables more faithful and generalisable reward recovery, especially in data-limited regimes.

We conduct our study in gridworld environments that simulate video game-like navigation tasks. These environments are compact yet allow us to introduce stochastic transitions, confounding dynamics, and variations in reward sparsity. We implement and compare three standard IRL baselines: the projection method by Ng and Russell [9], Maximum Entropy IRL [14], and Adversarial IRL (AIRL) [6]. These are evaluated on their ability to recover reward functions and imitate expert policies across different environment settings.

To extend beyond these baselines, we design a novel causal variant of AIRL that integrates a simplified causal model over expert state-action trajectories. Unlike standard AIRL, which treats transitions as purely Markovian, our method conditions on latent variables that reflect plausible real-world confounders (e.g., risk, intent, unobserved goals). We empirically evaluate the causal variant’s robustness in low-data scenarios and its ability to generalise to unseen or perturbed environments.

Our contributions are as follows:

- We implement and benchmark three canonical IRL methods under a comprehensive ablation suite covering discount factors, reward types, data regimes, and environment stochasticity.
- We propose a Causal-AIRL variant that encodes structural priors over expert behaviour and test its performance against baselines in confounded environments.
- We evaluate methods using both reward-level and trajectory-level metrics, including policy overlap, generalisation to held-out states, and reward alignment scores.

Ultimately, this work aims to bridge the gap between IRL and causality research, providing empirical evidence that causal reasoning enhances reward learning. In doing so, we contribute toward more robust and human-aligned agents³, particularly

³Human alignment refers to the agent’s behaviour matching human values, goals, or intentions, especially in ambiguous or underspecified tasks [8].

in safety-critical or interactive domains where imitation alone may fail to capture true intent [7, 3, 8].

Chapter 2

Related Work

2.1 Classical Inverse Reinforcement Learning

The original formulation of Inverse Reinforcement Learning (IRL) by Ng and Russell [9] introduced the problem of recovering a reward function from expert demonstrations. Their projection-based method aimed to find a reward under which the expert policy is optimal. However, the solution space is often underdetermined, leading to ambiguity in reward inference. Subsequent work, such as Maximum Margin Planning [11], attempted to regularise this space by imposing large-margin separation between expert and non-expert trajectories.

Ziebart et al.’s Maximum Entropy IRL (MaxEnt IRL) [14] addressed this ambiguity by framing reward inference as a probabilistic problem, seeking the distribution over trajectories with maximum entropy¹ subject to matching expert feature expectations. MaxEnt IRL became a cornerstone for later extensions into stochastic environments and deep learning-based models.

2.2 Deep and Adversarial IRL

The advent of deep learning enabled IRL methods to scale beyond tabular environments. Finn et al. [5] introduced Guided Cost Learning, which combined MaxEnt IRL with deep neural networks for reward representation. Building on this, Fu et al.’s Adversarial IRL (AIRL) [6] recast reward learning as a generative adversarial game. AIRL jointly learns a reward and policy by training a discriminator to distinguish expert from agent

¹The principle of maximum entropy favors the most uniform distribution over trajectories that still matches observed statistics, thereby reducing bias in reward inference [14].

trajectories.² Unlike earlier methods, AIRL can recover reward functions that are portable across environments, provided that the reward depends only on states.

While these advances improved reward learning fidelity and generalisation, they often assumed fully observable environments and ignored latent confounding factors that may influence demonstrated behaviour.

2.3 Causality in Imitation and Reinforcement Learning

Causal reasoning has recently been proposed as a way to improve policy generalisation and robustness. De Haan et al. [4] highlighted how standard imitation learning may learn spurious correlations when the demonstrator’s actions are influenced by hidden variables. Bica et al. [2] formalised this concern and proposed invariant causal imitation learning (ICIL), which leverages domain-invariant mechanisms across environments to learn robust policies.³

In parallel, Ruan et al. [12] extended the notion of causality to inverse reinforcement learning, arguing that causal knowledge can aid reward recovery under unobserved confounding. Zhang and Bareinboim [13] also introduced causal approaches for transfer learning in bandit and RL settings, showing improved generalisation under distribution shifts.

While these works establish the importance of causality, they stop short of integrating causal models directly into deep IRL frameworks like AIRL. This gap motivates our development of a Causal-AIRL variant that leverages structural priors to guide reward learning.

Causal-GAIL [4] extends the GAIL framework to learn policies that generalise across distributional shifts by enforcing invariance to latent confounders. This is achieved by conditioning the discriminator on environment-specific factors and learning domain-invariant representations. While effective for imitation under causal perturbations, Causal-GAIL does not explicitly recover a reward function, and thus cannot support reward-level interpretation or transfer. Our work, by contrast, focuses on causal reward inference: we extend Adversarial Inverse Reinforcement Learning (AIRL) [6] with latent structural priors to disentangle reward from confounded behaviour. This enables the recovery of a reward signal that is stable across expert variation and applicable

²AIRL uses a discriminator to shape the reward by distinguishing expert vs agent transitions, drawing from GAN-style training. This implicitly aligns reward learning with policy improvement [6].

³ICIL assumes that causal mechanisms remain stable across environments, enabling transfer of policy components that are causally correct [2].

beyond the demonstrated trajectories.

2.4 Human Alignment and Preference Learning

A parallel line of research focuses on learning reward functions that reflect human preferences. Hadfield-Menell et al. [7] introduced Cooperative IRL, where a human and robot cooperate to infer rewards. Christiano et al. [3] proposed learning from human preferences via pairwise trajectory comparisons, while Leike et al. [8] framed alignment as a broader reward modeling problem. These approaches share our goal of recovering human-aligned behaviours, though they often rely on additional supervision beyond demonstrations.

2.5 Summary

Our work builds on the intersection of these threads: we adopt the adversarial structure of AIRL, enrich it with causal assumptions inspired by recent work in imitation learning, and target generalisation and alignment as key evaluation axes. By benchmarking both classical and causal variants of IRL under limited data and perturbed conditions, we aim to illuminate the role of structural assumptions in reward inference.

Chapter 3

Background & Preliminaries

This chapter introduces the foundational concepts used throughout this dissertation. We review the formulation of Markov¹ Decision Processes, standard Inverse Reinforcement Learning (IRL), Adversarial IRL (AIRL), and causal inference tools relevant to this work.

3.1 Markov Decision Processes

We model the environment as a Markov Decision Process (MDP) defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, T, R, \gamma)$, where:

- $\mathcal{S}$ is the set of states,
- $\mathcal{A}$ is the set of actions,
- $T(s'|s, a)$ is the transition probability to state s' after taking action a in state s ,
- $R(s, a)$ is the reward function (unknown in IRL),
- $\gamma \in [0, 1)$ is the discount factor.

A stochastic policy $\pi(a|s)$ defines the agent's behaviour. The state value function $V^\pi(s)$ and action-value function $Q^\pi(s, a)$ quantify expected cumulative rewards:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s \right], \quad Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s, a_0 = a \right].$$

¹The Markov assumption posits that the next state depends only on the current state and action, not on past history. In confounded settings, this assumption may be violated.

3.2 Inverse Reinforcement Learning

In IRL, we are given expert demonstrations $\mathcal{D}_E = \{\tau_1, \dots, \tau_N\}$, where each trajectory τ_i is a sequence of state-action pairs sampled from an unknown expert policy π_E . The goal is to recover a reward function R under which π_E is (near-)optimal.

The problem is inherently ill-posed: many reward functions can explain the same behaviour. Classical approaches regularise the solution space by imposing maximum margin [11] or maximum entropy principles [14]. The latter leads to a probabilistic formulation where the expert’s trajectory distribution is modeled as:

$$P(\tau \mid R) \propto \exp \left(\sum_{t=0}^T R(s_t, a_t) \right).$$

3.3 Adversarial Inverse Reinforcement Learning

Adversarial IRL (AIRL) [6] casts IRL as a two-player game between a policy generator and a discriminator. AIRL assumes a parametric reward function $r_\theta(s, a)$ and approximates the occupancy measures of the expert and the learner via rollouts.² A discriminator $D(s, a)$ is trained to distinguish expert from learner trajectories, and its output is used to shape the reward:

$$r(s, a) = \log D(s, a) - \log(1 - D(s, a)).$$

The policy is updated via reinforcement learning on this learned reward.

Crucially, AIRL’s structure allows disentangling reward from dynamics by assuming the reward is a function of states alone (or (s, a)), and stationary across environments. This makes AIRL suitable for transfer and generalisation—key properties we wish to retain in our causal variant.

3.4 Causal Inference and Structural Models

To incorporate causal reasoning into IRL, we draw from the framework of Structural Causal Models (SCMs) [10]. An SCM represents variables as nodes in a directed acyclic graph (DAG), with structural equations describing dependencies:

$$X_i = f_i(\text{Pa}(X_i), U_i),$$

²The occupancy measure $\rho^\pi(s, a)$ gives the discounted visitation frequency of state-action pairs under policy π . It forms the core of IRL objectives by enabling feature expectation matching.

where $\text{Pa}(X_i)$ are the parents of X_i and U_i are exogenous noise variables.

A confounder is a variable that affects both the action and the reward but is not observed directly.³ In standard IRL, failure to model such latent causes can lead to spurious reward recovery [4, 12]. For example, if a demonstrator avoids a region due to risk (unobserved), the learner might attribute this to low immediate reward instead of strategic avoidance.

Causal IRL seeks to infer rewards while accounting for such structural dependencies. Methods like Invariant Causal Imitation Learning [2] use environment variations to identify invariant mechanisms. In our work, we explore how injecting a simplified causal graph over trajectories can improve AIRL’s robustness to confounding.

3.5 Notation Summary

For clarity, we summarise the key notations used throughout the dissertation in Table 3.1.

Symbol	Meaning
$\mathcal{S}, \mathcal{A}$	State and action spaces
$T(s' s, a)$	Transition probability
$R(s, a)$	Reward function (to be learned)
$\pi(a s)$	Stochastic policy
τ	Trajectory (sequence of (s_t, a_t) pairs)
$\mathcal{D}_E$	Set of expert demonstrations
$V^\pi(s), Q^\pi(s, a)$	Value and Q-functions
$D(s, a)$	Discriminator in AIRL
U	Unobserved confounders

Table 3.1: Summary of notation.

³A confounder is a hidden variable that affects both the expert’s action choice and the environment’s reward outcomes, introducing spurious correlations.

Chapter 4

Methods

This chapter details the algorithms and modeling assumptions used throughout the dissertation. We begin by establishing classical IRL baselines to ground our methodology in well-understood principles. We then build on these foundations to describe Adversarial Inverse Reinforcement Learning (AIRL) and introduce a novel causal extension. Each algorithm is evaluated under common conditions to ensure fair comparison, and care is taken to isolate the effects of confounding, reward sparsity, and limited demonstration data.

4.1 Classical IRL Baselines

4.1.1 Ng and Russell Projection Method

Ng and Russell [9] proposed one of the earliest IRL algorithms, designed to recover a reward function $R(s)$ under which the expert policy π_E is optimal. The method constructs a set of linear constraints that encode the condition:

$$\sum_s \mu_E(s) R(s) \geq \sum_s \mu^\pi(s) R(s), \quad \forall \pi \in \Pi,$$

where $\mu_E(s)$ is the expert’s state visitation frequency and $\mu^\pi(s)$ is that of a competing policy π . The algorithm solves for a reward R that maximizes the margin between the expert and the next-best policy, subject to these constraints.

While interpretable and efficient in small environments, the projection method suffers from ambiguity in the reward (non-uniqueness) and requires full knowledge of environment dynamics to enumerate all π .

4.1.2 Maximum Entropy IRL

Ziebart et al. [14] introduced a probabilistic formulation that resolves reward ambiguity by assuming the expert follows the maximum entropy distribution over trajectories that matches feature expectations. The expert trajectory distribution is modeled as:

$$P(\tau) \propto \exp \left(\sum_{t=0}^T R(s_t, a_t) \right).$$

The reward is typically linear in hand-crafted features $\phi(s, a)$, i.e., $R(s, a) = \theta^\top \phi(s, a)$, and the algorithm optimizes:

$$\min_{\theta} \text{KL}(\rho_E \parallel \rho_{\theta}),$$

where ρ_E and ρ_{θ} are the feature expectations under the expert and the learned policy, respectively.

MaxEnt IRL is robust to stochasticity and can model suboptimal expert behaviour more gracefully than classical methods, but requires accurate estimation of partition functions and feature expectations, which can be computationally demanding.

4.1.3 Baseline Evaluation Criteria

The classical IRL baselines are evaluated using:

- **Reward correlation:** Pearson correlation between recovered and ground truth reward across the state space.
- **Value function distance:** ℓ_2 norm between V^{π_E} and $V^{\pi_{\text{learned}}}$.
- **Policy agreement:** proportion of states where the learned policy matches expert actions.
- **Reward map visualisation:** heatmaps of recovered vs. true reward to expose spatial structure.

These metrics allow us to benchmark performance under variable conditions: known vs. unknown γ , sparse vs. shaped rewards, and varying numbers of demonstrations.

4.2 Adversarial Inverse Reinforcement Learning

Building on classical IRL, AIRL [6] uses an adversarial framework to learn a reward function from expert demonstrations via a discriminator trained to distinguish between expert and learner trajectories.

4.2.1 Learning Objective

The discriminator is parameterized as:

$$D_{\theta}(s, a) = \frac{\exp(f_{\theta}(s, a))}{\exp(f_{\theta}(s, a)) + \pi_{\phi}(a|s)},$$

and is trained via the binary cross-entropy loss:

$$\mathcal{L}_D(\theta) = \mathbb{E}_{(s,a) \sim \mathcal{D}_E} [\log D_{\theta}(s, a)] + \mathbb{E}_{(s,a) \sim \mathcal{D}_{\pi}} [\log(1 - D_{\theta}(s, a))].$$

The reward used for RL updates is given by:

$$r(s, a) = \log D_{\theta}(s, a) - \log(1 - D_{\theta}(s, a)).$$

4.2.2 Reward Structure

To ensure the reward generalises across dynamics, f_{θ} is often decomposed as:

$$f_{\theta}(s, a, s') = r_{\theta}(s, a) + \gamma h_{\theta}(s') - h_{\theta}(s),$$

which recovers a potential-based shaping function h_{θ} . This structure allows AIRL to disentangle reward from the policy-induced visitation distribution.

4.2.3 Limitations

AIRL assumes:

- Fully observable states and stationary dynamics,
- Sufficiently large expert dataset to match occupancy measures,
- No hidden confounders influencing expert behaviour.

These assumptions break down in low-data regimes or environments with latent structure, leading to brittle or misaligned rewards.

4.3 Causal Adversarial Inverse Reinforcement Learning

To address these issues, we propose *Causal-AIRL*: a causal extension of AIRL that explicitly models latent variables Z mediating the relationship between states, actions, and optimality.

4.3.1 Latent Structural Model

We posit a causal graph of the form:

$$Z \rightarrow A \rightarrow S', \quad Z \rightarrow S',$$

where Z captures unobserved intent or strategic factors. Expert actions are drawn from $\pi_E(a|s, Z)$, and transitions from $T(s'|s, a, Z)$. The goal is to recover a reward $r(s, a)$ invariant to Z .

4.3.2 Discriminator Conditioning and Inference

We augment the discriminator to be conditioned on z , either inferred or marginalised:

$$D_\theta(s, a) = \frac{\exp(f_\theta(s, a, z))}{\exp(f_\theta(s, a, z)) + \pi_\phi(a|s)},$$

with z sampled from an inference model $q_\phi(z|s, a)$. The training objective becomes a variational lower bound:

$$\log P(s, a) \geq \mathbb{E}_{z \sim q_\phi(z|s, a)} [\log D_\theta(s, a, z)] - \text{KL}[q_\phi(z|s, a) \parallel P_\Psi(z)],$$

where $P_\Psi(z)$ is a learned prior over latent variables.

4.3.3 Training Loop

The full training pipeline alternates:

1. Generate agent rollouts $\mathcal{D}_\pi$.
2. Update discriminator D_θ with (s, a, z) from expert and agent data.
3. Update inference model $q_\phi(z|s, a)$ and prior $P_\Psi(z)$.
4. Update policy π_ϕ via RL using the learned reward.

This process aims to isolate causal reward signals from observational biases introduced by Z .

4.4 Environment Design and Simplifications

To ensure interpretability and tractable evaluation, we use tabular gridworld environments with full observability and known transition dynamics. This setting enables us to:

- Conduct reward recovery under controlled perturbations,
- Explicitly test generalisation to unvisited or perturbed states,
- Vary reward structure (terminal-only vs. shaped, positive vs. negative),
- Vary demonstration quality and number,
- Evaluate robustness to misspecified γ and transitions T .

These simplifications allow us to systematically test the role of causal assumptions, reward sparsity, and structural generalisation in reward learning.

In high-dimensional continuous environments such as CartPole, causal structure can be formalised using structural causal models (SCMs). Specifically, environment dynamics can be influenced by latent confounders such as pole length, cart mass, or friction — which affect both the agent’s observations and the outcomes of its actions. For instance, different pole lengths can alter the optimal policy even if the task objective remains unchanged. These confounders, if not explicitly accounted for, can lead to spurious correlations between observed states and rewards.

Causal-AIRL addresses this by introducing a latent variable Z into the discriminator-reward learning pipeline, allowing the model to learn rewards that remain invariant across variations in environment dynamics. In the SCM view, Z captures exogenous factors that influence state transitions and rewards, formalised as $Z \rightarrow s_t \rightarrow a_t \rightarrow s_{t+1}$, with feedback from Z to s_{t+1} . By marginalising over Z or conditioning reward learning on inferred values, we seek to recover a stable reward function that generalises across different structural configurations of the environment.

Chapter 5

Experiments

This chapter presents a comprehensive evaluation of the IRL algorithms introduced in Chapter 4. We begin with a description of the experimental setup, including environment design, reward structures, and demonstration protocols. We then define the metrics used to evaluate reward recovery and policy fidelity, followed by ablation studies and generalisation tests. The goal is to systematically assess each method’s robustness to sparse data, confounding, reward design, and environmental perturbations.

5.1 Experimental Setup

5.1.1 Environments

All experiments are conducted in tabular gridworld environments designed to simulate video game-like navigation tasks. These environments are deterministic or stochastic variants of finite-state MDPs with known transitions and full observability. States represent positions in a 2D grid, and actions include cardinal movements (up, down, left, right). We use both fixed layouts and procedurally generated variations to test generalisation.

To model environment stochasticity, we include a `slippery` flag that induces random action noise, i.e., with probability p , the intended action is replaced by a random alternative.

5.1.2 Extension to Continuous Environments: CartPole

While tabular environments such as GridWorld provide interpretability and control, they may not fully reflect the capabilities of deep IRL methods. To evaluate generalisability

beyond discrete state spaces, we additionally apply AIRL and Causal-AIRL to the CartPole-v1 environment, a continuous control benchmark commonly used in imitation learning.

CartPole presents a partially observable Markov decision process (POMDP) with continuous state features (e.g., pole angle, cart velocity) and binary actions. It provides a natural testbed for evaluating the learned reward’s ability to guide balance under dynamic conditions.

We exclude MaxEnt IRL from this environment, as it is designed for tabular domains with discrete feature counts and lacks the deep policy optimiser required to scale to continuous spaces.

For AIRL and Causal-AIRL, we train agents using expert trajectories collected from a near-optimal policy. The reward function is learned using the standard discriminator framework, and a soft actor-critic (SAC) policy is trained on the inferred reward. Evaluation is based on cumulative return, reward visualisation, and policy stability under perturbed dynamics.

In the Causal-AIRL setting, we introduce a latent confounder Z to simulate causal variation — such as unobserved differences in pole length or cart mass — and evaluate whether the inferred reward remains invariant across such shifts. This allows us to test the causal robustness and generalisability of the learned reward in more realistic, high-dimensional settings.

5.1.3 Reward Structures

We evaluate reward learning under the following configurations:

- **Sparse rewards:** Non-zero reward is given only at terminal states.
- **Shaped rewards:** Intermediate rewards guide progress toward goals.
- **Positive vs. negative rewards:** Rewards indicate desirability or penalties respectively.
- **Intermediate vs. terminal-only:** To test the recoverability of temporally extended credit assignment.

These variations help isolate how the expressivity and density of the reward signal affect recovery and policy learning.

5.1.4 Expert Demonstrations

Expert demonstrations are generated via soft-optimal policies computed using dynamic programming (for small environments) or entropy-regularised value iteration. We vary:

- Number of expert trajectories $N \in \{1, 2, 5, 10, 20, 30, 50\}$.
- Demonstrator optimality: perfect expert vs. noisy/stochastic expert.
- Observability of latent confounders Z (in causal experiments).

In causal settings, we simulate experts with different latent preferences (e.g., risk-averse vs. risk-seeking) and test whether reward learners can infer a consistent reward signal despite this variation.

5.1.5 Implementation Details

Classical IRL methods (Ng-Russell, MaxEnt) are implemented in tabular form with exact feature counts. AIRL and Causal-AIRL use tabular approximators for r_θ and D_θ , trained using stochastic gradient descent.

Policy optimisation is performed using soft Q-learning or entropy-regularised policy iteration. Unless specified, we assume the discount factor γ is known. In ablation settings, we test recovery under γ misspecification.

All experiments are run with fixed random seeds. Hyperparameters (e.g., learning rate, batch size) are selected via small-scale grid search to ensure convergence.

5.2 Evaluation Metrics

5.2.1 Reward-Level Metrics

- **Reward correlation:** Pearson correlation between recovered and true reward across the state space.
- **Mean squared error (MSE):** $\frac{1}{|S|} \sum_s (R_{\text{true}}(s) - R_{\text{learned}}(s))^2$.
- **Reward generalisation:** Evaluation of R_{learned} on perturbed or held-out states.

5.2.2 Policy-Level Metrics

- **Policy agreement:** Proportion of states where π_{learned} matches π_E .
- **Value difference:** $\|V^{\pi_E} - V^{\pi_{\text{learned}}}\|_2$.
- **Trajectory overlap:** Jaccard similarity or edit distance between expert and learned trajectories.

5.2.3 Causal Metrics

- **Reward invariance:** Variance of recovered reward under different expert latent profiles Z .
- **Confounder robustness:** Reward quality when demonstrations are confounded vs unconfounded.

5.2.4 Qualitative Analysis

Visualisation tools include:

- Heatmaps of R_{learned} vs. R_{true} .
- Vector fields showing reward gradients or policy preferences.
- Trajectory plots comparing expert and learned rollouts.

5.3 Parameter Sensitivity Analysis

This section investigates how the behaviour of IRL algorithms varies across key structural parameters. Rather than treating this as a classical ablation study — which typically involves removing or altering model components — we frame this as a structured parameter sensitivity analysis. Each sweep explores the effect of a single variable on reward recovery, policy fidelity, and robustness, while controlling for other factors. The goal is to understand the empirical and causal behaviour of each algorithm under challenging conditions.

Effect of Discount Factor γ

We vary $\gamma \in \{0.8, 0.9, 0.99\}$ to analyse the influence of planning horizon. Lower γ prioritises short-term rewards and can lead to myopic behaviour, while higher γ encourages long-term planning. AIRL and Causal-AIRL in particular are known to be sensitive to γ due to their policy gradient backbone. We evaluate reward correlation, value difference, and policy agreement under these settings.

Effect of Demonstration Count N

We sweep over $N \in \{1, 2, 3, 5, 8, 10, 15, 20, 30, 50\}$ to test data efficiency and convergence stability. Sparse settings test few-shot generalisation, while larger N assess asymptotic performance. This is particularly important for evaluating causal variants under limited demonstrations.

Effect of Softmax Temperature T

We examine softmax temperature $T \in \{0.1, 0.5, 1.0\}$ in the probabilistic policy evaluation step. Lower T sharpens the policy (making it more deterministic), while higher T induces exploration and smooths the reward landscape. This parameter directly affects the learned policy and indirectly impacts reward estimation when using feature expectations.

Effect of Reward Type

We test reward types `sparse`, `shaped`, `negative`, and `positive` to assess algorithmic sensitivity to different supervision signals. Sparse rewards evaluate generalisation under weak guidance. Shaped rewards provide potential-based heuristics. Positive/negative rewards isolate the impact of reward sign and magnitude on learning dynamics. Understanding this helps test whether causal variants learn intent rather than surface features.

Effect of Confounding (Z On/Off)

We compare model performance on environments with and without confounding. This tests the core hypothesis of causal IRL — that explicitly modelling latent structure (via Z) allows for more robust reward recovery. Causal-AIRL should ideally show lower reward variance and better generalisation across expert types.

Effect of Transition Noise

To evaluate robustness under environment stochasticity, we repeat experiments in the “slippery” version of the GridWorld environment where transitions are non-deterministic. This tests whether methods can recover consistent rewards and stable policies despite noise in state transitions, which is a common feature of real-world dynamics.

5.3.1 Sanity Checks and Implementation Validation

To ensure the correctness of our classical IRL implementations, we evaluate both the Ng-Russell and MaxEnt IRL methods on small, analytically solvable environments. Specifically, we construct 3×3 and 4×4 gridworlds with deterministic transitions and a single goal state. The expert demonstrations are generated from a known optimal policy under a handcrafted reward function.

These environments allow us to verify whether the recovered reward:

- Is highest near the goal state (for positive reward tasks),
- Matches the known reward structure under feature expectations (MaxEnt IRL),
- Produces a learned policy that aligns with the expert trajectory distribution.

We also verify that the recovered value functions are monotonic with respect to distance to the goal, and that the learned policy matches the expected action distribution under the original reward. These experiments act as functional baselines to validate implementation correctness before evaluating on larger or confounded environments.

To support reproducibility, we include scripted test cases under `tests/` that execute these gridworld sanity checks automatically.

5.4 Generalisation Tests

5.4.1 Held-Out States

We remove portions of the state space during training and test if learned policies generalise sensibly to these regions.

5.4.2 Environment Perturbations

We modify layouts or transition kernels after training and evaluate whether the learned reward leads to robust, adaptive behaviour.

5.4.3 Cross-Policy Learning

We train reward models on suboptimal expert demonstrations or multiple expert types and test if the learned reward captures an invariant core across demonstrations.

5.5 Implementation Summary

- All models are trained on CPU with runtime under 10 minutes per experiment.
- Ng-Russell and MaxEnt complete within seconds; AIRL and Causal-AIRL require more iterations but converge reliably.
- Code is modular, supporting plug-and-play reward types, confounders, and trajectory samplers.
- Training stability is achieved via entropy regularisation and early stopping on held-out transitions.

This controlled and interpretable setup allows us to systematically study the influence of causal structure, reward design, and data availability on inverse reinforcement learning outcomes.

Chapter 6

Results & Discussion

This chapter presents and interprets the empirical findings of our experiments. We analyse the performance of classical IRL baselines, AIRL, and our proposed Causal-AIRL across a series of controlled settings. The evaluation focuses on three core aspects: (i) reward recovery accuracy, (ii) policy fidelity, and (iii) generalisation under structural variation. We draw conclusions from both quantitative metrics and qualitative visualisations.

6.1 Overview

Our experiments address the following research questions:

- How do IRL algorithms perform under varying data availability and reward structures?
- Can causal assumptions improve reward recovery when expert demonstrations are confounded?
- Do learned rewards generalise to unseen states or perturbed environments?

Results are presented through performance curves, heatmaps, and policy rollouts. Each section highlights the conditions under which specific methods excel or fail.

6.2 Classical IRL Baseline Results

Figure ?? shows the reward correlation and value difference achieved by Ng-Russell and MaxEnt IRL across varying demonstration counts. MaxEnt consistently outperforms

the projection method under sparse rewards, particularly for small N . This aligns with its entropy-based formulation, which better tolerates suboptimality and data sparsity.

Heatmaps in Figure ?? demonstrate that MaxEnt more accurately recovers the spatial structure of reward, while the projection method often collapses to degenerate solutions in low-data settings. Both methods are sensitive to shaped vs sparse reward variants, with MaxEnt showing improved gradient propagation toward intermediate goals.

Overall, classical methods serve as useful references, but struggle with generalisation and reward ambiguity when demonstrations are limited.

6.3 AIRL Performance Analysis

AIRL demonstrates improved scalability and reward fidelity compared to classical methods, particularly in shaped reward environments. As shown in Figure ??, AIRL achieves higher policy agreement and lower value difference than MaxEnt IRL beyond $N \geq 5$.

However, AIRL’s performance degrades when demonstrations are sparse and confounded. Figure ?? shows cases where the discriminator overfits to behavioural artefacts rather than underlying intent. Reward maps become fragmented, and learned policies fail to generalise outside the training regime.

We also observe that AIRL is sensitive to the reward shaping assumption encoded in its discriminator structure. In some cases, this leads to spurious gradients in regions where the expert provided no information.

6.4 Causal-AIRL Results

Causal-AIRL demonstrates consistent improvements in settings with latent confounding. Figure ?? reports reward variance across different expert types (varying Z), showing that Causal-AIRL recovers an invariant reward, while AIRL adapts to each demonstrator’s idiosyncrasies.

Figure ?? compares policy rollouts from AIRL and Causal-AIRL trained on confounded data. Only Causal-AIRL generalises to the intended goal states across all latent conditions. This supports our hypothesis that causal conditioning enables better disentanglement of intent from surface behaviour.

When Z is marginalised or inferred from context, reward gradients remain coherent and generalisable. Training remains stable, and convergence speed is comparable to AIRL despite the additional latent inference step.

6.5 Ablation Insights

6.5.1 Demonstration Count

Figure ?? shows that all methods benefit from increased N , but Causal-AIRL outperforms others in low-data settings. Notably, Causal-AIRL matches AIRL’s peak performance with half the demonstrations.

6.5.2 Reward Type Sensitivity

Figure ?? compares performance under sparse vs. shaped rewards. AIRL and Causal-AIRL handle shaped rewards effectively, but only the latter maintains stable gradients under sparse conditions. MaxEnt IRL performs well when shaped rewards are used, but suffers from distributional shift under evaluation.

6.5.3 Confounding Robustness

Figure ?? highlights that AIRL’s reward changes significantly across expert types, indicating overfitting to spurious features. Causal-AIRL exhibits minimal reward drift, suggesting it captures structure invariant to the confounder Z .

6.5.4 Transition Noise Robustness

Under `slippery` dynamics, AIRL and Causal-AIRL degrade gracefully, whereas classical methods struggle. Causal-AIRL exhibits lower variance across seeds, likely due to its use of structural latent conditioning.

6.5.5 Discount Factor Sensitivity

Figure ?? reports performance when γ is misspecified during training. AIRL is highly sensitive to incorrect γ , while Causal-AIRL shows robustness up to a 10% deviation. This suggests that encoding causal structure mitigates the effect of planning horizon misspecification.

6.6 Generalisation Performance

6.6.1 Held-Out States

Table ?? summarises performance on unseen regions of the state space. Causal-AIRL exhibits the best generalisation, maintaining coherent policy behaviour without retraining. AIRL and MaxEnt tend to overfit to the training distribution.

6.6.2 Environment Perturbations

Under layout and transition changes (Figure ??), Causal-AIRL consistently generalises better than AIRL. Rewards learned by AIRL often become brittle, while Causal-AIRL retains directional consistency.

6.6.3 Cross-Policy Learning

When trained on demonstrations from multiple expert types, only Causal-AIRL infers a reward consistent with all sources (Figure ??), suggesting that its inductive bias supports aggregation of diverse inputs.

6.7 Qualitative Analysis

Figure ?? displays selected heatmaps, vector fields, and trajectory plots for each method. Key insights include:

- Causal-AIRL learns smoother and more interpretable reward landscapes.
- AIRL reward fields often contain discontinuities when trained on heterogeneous demonstrations.
- Classical methods collapse under confounding or sparse supervision.

Training curves (Appendix ??) show stable convergence for AIRL and Causal-AIRL, with lower variance in the latter.

6.8 Discussion

Our findings confirm that:

- Classical IRL methods provide fast but limited reward estimates.
- AIRL scales to complex distributions but suffers under confounding.
- Causal-AIRL improves robustness, generalisation, and interpretability under minimal assumptions.

These results demonstrate that causal structure enhances reward learning, particularly when data is limited or inconsistent. The benefits come at the cost of added modeling complexity, but the gains in policy fidelity and reward stability justify this extension.

Limitations

While our framework is effective in controlled settings, several limitations remain:

- Environments are fully observable and low-dimensional.
- Latent confounders are simulated rather than inferred from human data.
- The causal model is simplified and manually specified.

Future Work

Future directions include:

- Extending to partial observability and high-dimensional observation spaces.
- Integrating causal discovery for latent structure.
- Applying to real-world tasks with expert variation (e.g., surgical robotics, autonomous driving).

Our work provides an empirical foundation for causally grounded reward learning, demonstrating the potential of causal structure in IRL beyond imitation.

While our work focuses on learning invariant rewards from confounded demonstrations, an alternative line of research has explored causal robustness at the policy level. Methods such as Causal-GAIL aim to learn policies that generalise across interventions without explicitly recovering rewards. These approaches are complementary: future work could explore hybrid frameworks that integrate causal reward inference with policy-invariant regularisation [2], particularly in high-dimensional or partially

observed domains. This could yield agents that are both behaviourally robust and value-aligned.

Chapter 7

Conclusions

7.1 Final Reminder

The body of your dissertation, before the references and any appendices, *must* finish by page 40. The introduction, after preliminary material, should have started on page 1.

You may not change the dissertation format (e.g., reduce the font size, change the margins, or reduce the line spacing from the default 1.5 spacing). Be careful if you copy-paste packages into your document preamble from elsewhere. Some L^AT_EX packages, such as `fullpage` or `savetrees`, change the margins of your document. Do not include them!

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Appendix A

First appendix

A.1 First section

Any appendices, including any required ethics information, should be included after the references.

Markers do not have to consider appendices. Make sure that your contributions are made clear in the main body of the dissertation (within the page limit).